Experiment 9
Clustering: K-Means Algorithm

Aim:

To perform clustering using the K-Means algorithm and determine the optimal number of clusters using the Elbow method.

Theory :

Clustering is the task of grouping observations in such a way that members of the same cluster are more similar to each other and members of different clusters are very different from each other. Clustering is commonly used to explore a dataset to either identify the underlying patterns in it or to create a group of characteristics.

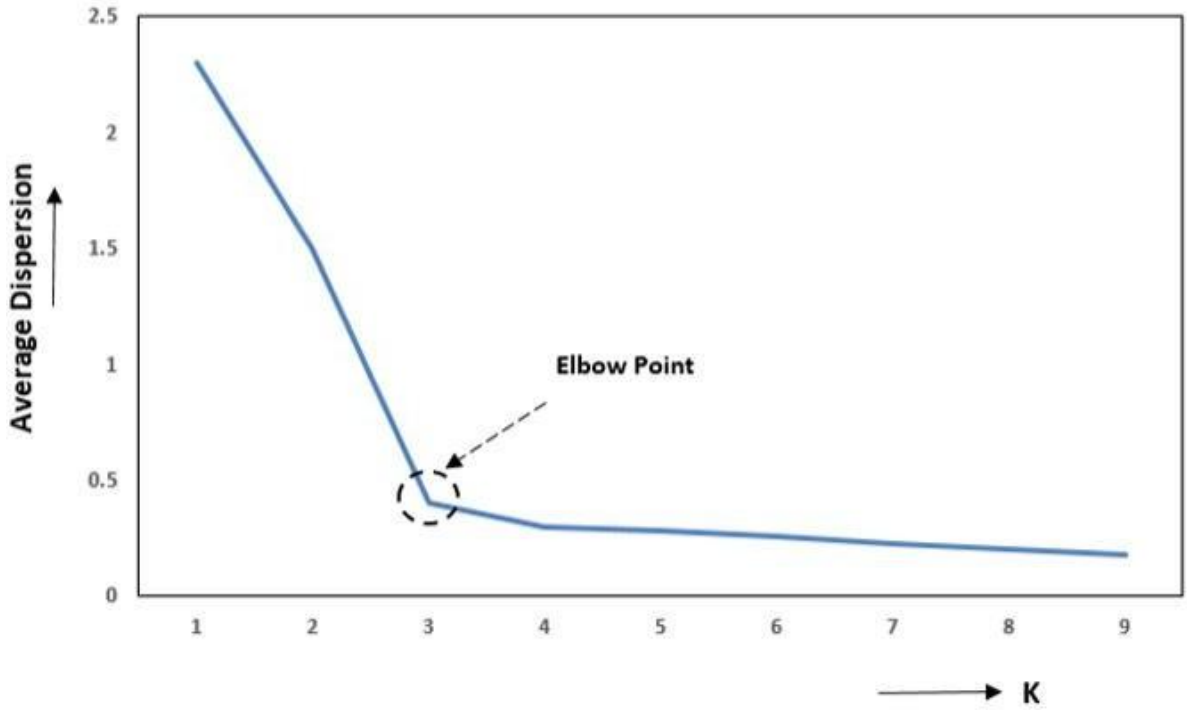
The k-means clustering algorithm is an iterative process of moving the centers of clusters or centroids to the mean position of their constituent points, and reassigning instances to their closest clusters iteratively until there is no significant change in the number of cluster centers possible or number of iterations reached.

The cost function of k-means is determined by the Euclidean distance (square-norm) between the observations belonging to that cluster with its respective centroid value. An intuitive way to understand the equation is, if there is only one cluster (k=1), then the distances between all the observations are compared with its single mean. Whereas, if, number of clusters increases to 2 (k= 2), then two-means are calculated and a few of the observations are assigned to cluster 1 and other observations are assigned to cluster two based on proximity. Subsequently, distances are calculated in cost functions by applying the same distance measure, but separately to their cluster centers:

Centroid distance calculations are performed by taking Euclidean distances. The Euclidean distance between two points A (X1, Y1) and B (X2, Y2) is shown as follows:

$$Euclidean\ distance\ between\ A\ \&\ B = \sqrt{(X2 - X1)^2 + (Y2 - Y1)\text{^}2}$$

To determine the optimal number of clusters, the Elbow method is used. The elbow method plots the value of the cost function produced by different values of k. The value of k at which improvement in distortion declines the most is called the elbow, at which we should stop dividing the data into further clusters. The following figure shows the Elbow method for selecting the optimal number of clusters. The elbow point indicates the most suitable number of clusters.



The elbow method is used to determine the optimal number of clusters in k-means clustering. The elbow method plots the value of the cost function produced by different values of k. The value of k at which improvement in distortion declines the most is called the elbow, at which we should stop dividing the data into further clusters.

Merits of K-Means clustering:

- K-Means is simple to understand and easy to implement, making it suitable for clustering large datasets.
- It is computationally efficient and scales well when the number of clusters and features is relatively small.
- The algorithm provides clear and compact cluster representations using centroids, which are easy to interpret.

Demerits of K-Means clustering:

- It is sensitive to the initial selection of centroids and may converge to a local optimum.
- The algorithm performs poorly with non-spherical clusters, varying cluster sizes, and is highly sensitive to outliers.

Algorithm

Step 1: Choose K (number of clusters)

Step 2: Initialize K centroids:

- **Random:** Pick K random data points as initial centroids
- **Pick first centroid randomly:** For each subsequent centroid, pick point with probability proportional to squared distance from nearest existing centroid

Step 3: REPEAT until convergence:

- **Step 3a: Assignment Step**
  - For each data point:
    1. Calculate distance to all K centroids
    2. Distance formula: $d = \sqrt{[(x_1-c_1)^2 + (x_2-c_2)^2 + ...]}$
    3. Assign point to cluster with nearest centroid
- **Step 3b: Update Step**
  - For each cluster k:
    1. Calculate new centroid as mean of all points in cluster
    2. New centroid coordinates = $(\Sigma x_i/n, \Sigma y_i/n, ...)$
- **Step 3c: Check Convergence**
  - If centroids haven't moved (or moved less than threshold) then **STOP**
  - If maximum iterations reached then **STOP**
  - Otherwise → Continue to Step 3a

Step 4: Output final cluster assignments and centroid positions

K-Means Clustering - Practical Implementation

Virtual Labs - Dayalbagh Educational Institute

Step 1: Importing Libraries

Importing Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
from sklearn.metrics import davies_bouldin_score
print("Libraries Imported")
```

Output:

```
Libraries Imported
```

Step 2: Loading Dataset

Reading Mall_Customers dataset

```
data = pd.read_csv("Mall_Customers.csv")
print("Dataset loaded successfully")
```

Output:

```
Dataset loaded successfully
```

Step 3: Data Analysis

Display the first 5 rows of the dataset

```
data.head()
```

Output:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

Display the last 5 rows of the dataset

```
data.tail()
```

Output:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
195	196	Female	35	120	79
196	197	Female	45	126	28
197	198	Male	32	126	74
198	199	Male	32	137	18
199	200	Male	30	137	83

Display the summary statistics of the dataset

```
data.describe()
```

Output:

	CustomerID	Age	Annual Income (k$)	Spending Score (1-100)
count	200.000000	200.000000	200.000000	200.000000
mean	100.500000	38.850000	60.560000	50.200000
std	57.879185	13.969007	26.264721	25.823522
min	1.000000	18.000000	15.000000	1.000000
25%	50.750000	28.750000	41.500000	34.750000
50%	100.500000	36.000000	61.500000	50.000000
75%	150.250000	49.000000	78.000000	73.000000
max	200.000000	70.000000	137.000000	99.000000

Display the information of the dataset

```
data.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
```

Output:

#	Column	Non-Null Count	Dtype
0	CustomerID	200 non-null	int64
1	Gender	200 non-null	object
2	Age	200 non-null	int64
3	Annual Income (k$)	200 non-null	int64
4	Spending Score (1-100)	200 non-null	int64

Output:

```
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

Check for missing values

```
data.isnull().sum()
```

Output:

Column	Missing Values
CustomerID	0
Gender	0
Age	0
Annual Income (k$)	0
Spending Score (1-100)	0

Output:
```
dtype: int64
```

```
# Display the shape of the dataset

data.shape
```

Output:
```
(200, 5)
```

Step 4: Data Preprocessing

```
# Convert the 'Gender' column from categorical values (Male/Female) to numeric labels (0/1) and preview the updated

data['Gender'] = data['Gender'].map({'Male': 0, 'Female': 1})
data.head()
```

Output:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	0	19	15	39
1	2	0	21	15	81
2	3	1	20	16	6
3	4	1	23	16	77
4	5	1	31	17	40

```
# Select Annual Income and Spending Score as feature variables for clustering and display a confirmation message

X = data[['Annual Income (k$)', 'Spending Score (1-100)']]
print("Features selected for clustering: Annual Income and Spending Score")
```

Output:
```
Features selected for clustering: Annual Income and Spending Score
```

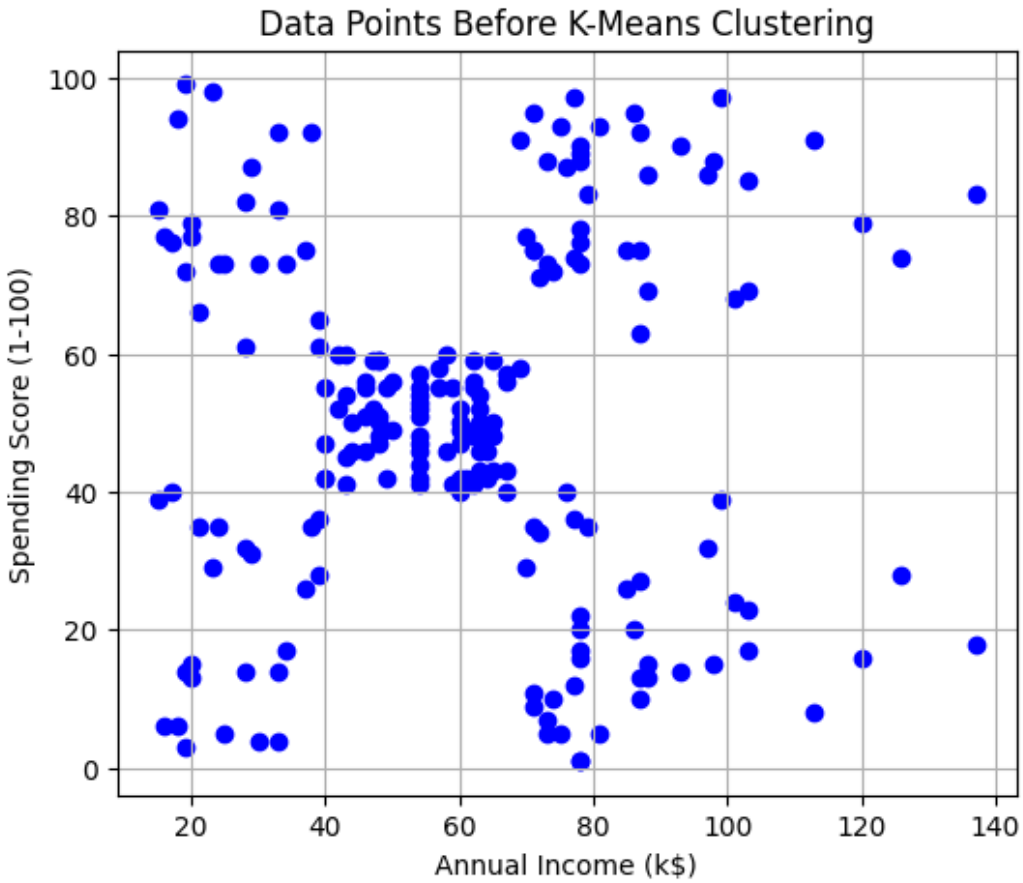
```
# Visualize the original data points by plotting Annual Income against Spending Score before applying K-Means clust
```

```
plt.figure(figsize=(6,5))
plt.scatter(
    X['Annual Income (k$)'],
    X['Spending Score (1-100)'],
    color='blue'
)

plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score (1-100)")
plt.title("Data Points Before K-Means Clustering")
plt.grid()
plt.show()
```

Output:

```
Data Points Before K-Means Clustering
```



Step 5: Model Training

 # Apply StandardScaler to normalize the selected features so they have zero mean and unit variance

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
print("Feature scaling complete using StandardScaler")
```

Output:

```
Feature scaling complete using StandardScaler
```

 # Compute WCSS for different numbers of clusters and plot the Elbow Curve to identify the optimal value of K

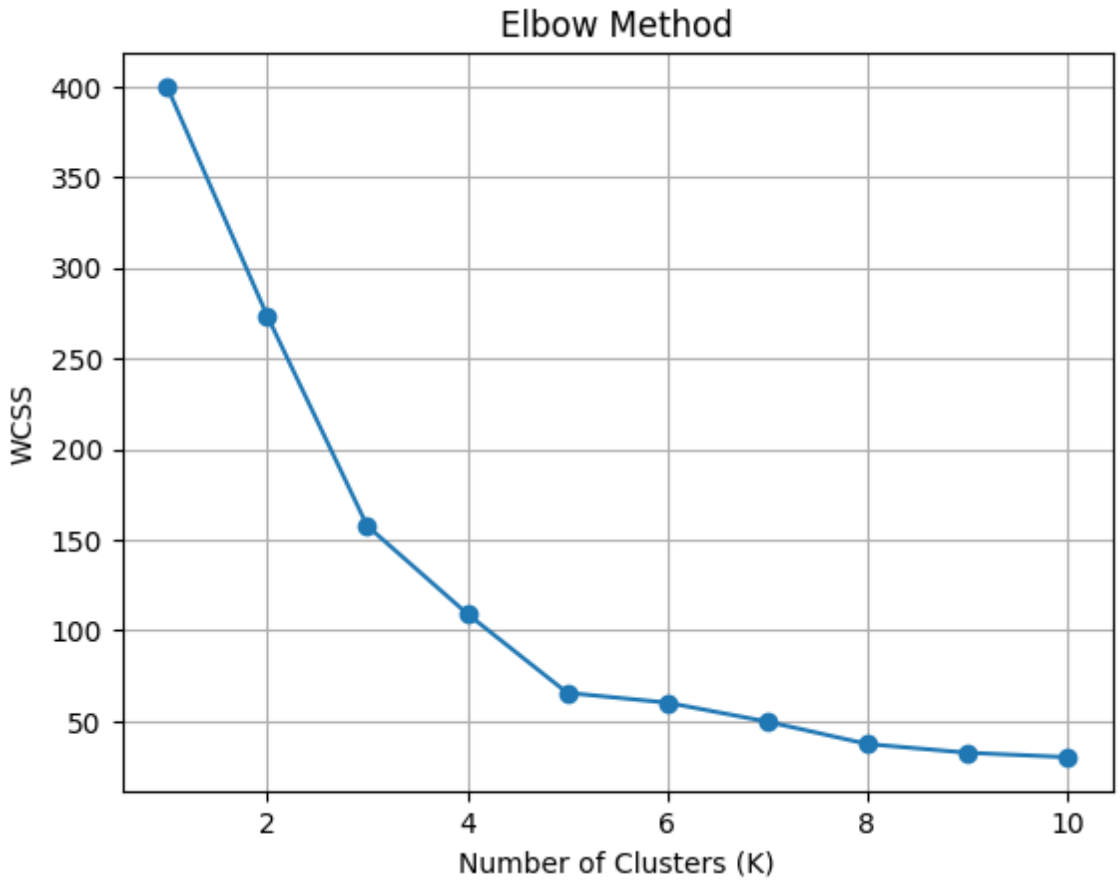
```
wcss = []

for k in range(1,11):
    kmeans = KMeans(n_clusters=k, random_state=42)
    kmeans.fit(X_scaled)
    wcss.append(kmeans.inertia_)

plt.plot(range(1,11), wcss, marker='o')
plt.xlabel("Number of Clusters (K)")
plt.ylabel("WCSS")
plt.title("Elbow Method")
plt.grid()
plt.show()
```

Output:

```
Elbow Method (k=5)
```



 # Initialize the K-Means algorithm with 5 clusters and train it on the scaled feature data

```
kmeans = KMeans(n_clusters=5, random_state=42)
kmeans.fit(X_scaled)
print("Model trained successfully")
```

Output:

```
Model trained successfully
```

```
KMeans
KMeans(n_clusters=5, random_state=42)
```

Step 6: Model Evaluation

 # Assign cluster labels to the original dataset and display the first 5 rows with the new 'Cluster' column

```
labels = kmeans.labels_
data['Cluster'] = labels
data[['Annual Income (k$)', 'Spending Score (1-100)', 'Cluster']].head()
```

Output:

	Annual Income (k$)	Spending Score (1-100)	Cluster
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

 # Visualize the clusters by plotting Annual Income against Spending Score, with colors representing cluster labels an

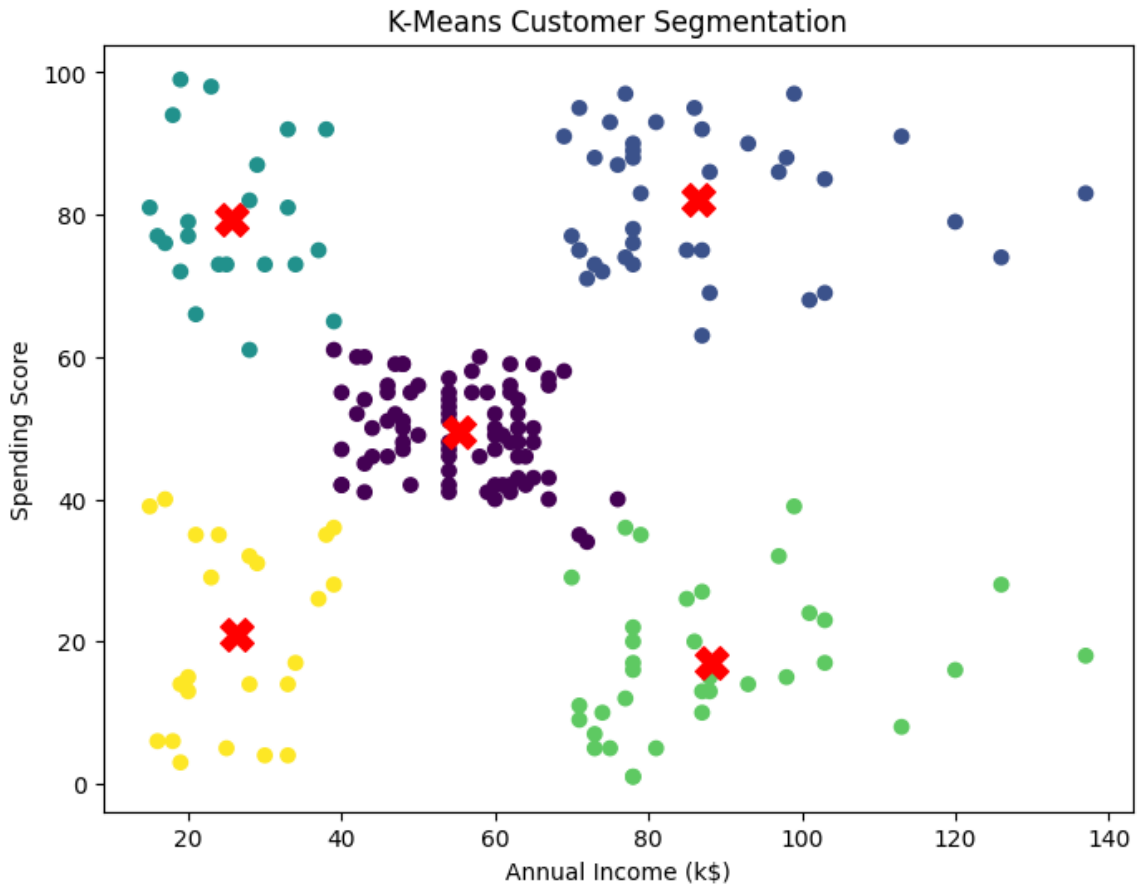
```
plt.figure(figsize=(8,6))
plt.scatter(
    X['Annual Income (k$)'],
    X['Spending Score (1-100)'],
    c=labels,
    cmap='viridis'
)

centers = scaler.inverse_transform(kmeans.cluster_centers_)
plt.scatter(
    centers[:,0],
    centers[:,1],
    c='red',
    s=200,
    marker='X'
)

plt.xlabel("Annual Income (k$)")
plt.ylabel("Spending Score")
plt.title("K-Means Customer Segmentation")
plt.show()
```

Output:

```
K-Means Customer Segmentation
```



 # Calculate and display model evaluation metrics: Silhouette Score, WCSS (Inertia), and Davies-Bouldin Index

```
score = silhouette_score(X_scaled, labels)
print("Silhouette Score:", score)

wcss = kmeans.inertia_
print("WCSS (Inertia):", wcss)

db_score = davies_bouldin_score(X_scaled, labels)
print("Davies-Bouldin Index:", db_score)
```

Output:

```
Silhouette Score: 0.5546571631111091
WCSS (Inertia): 65.56840815571681
Davies-Bouldin Index: 0.5722356162263352
```

Step 7: Model Simulation

 # Interactive K-Means Simulation

```
Click the 'Enter Interactive Animation' button to open the full-screen simulator.
Explore how K-Means Clustering segments the customer dataset in real-time.
```

Output:

```
Full-Screen Simulator Ready. Click the 'Enter Interactive Animation' button to start exploring.
```